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Processsafetyandabnormalsituationmanagement

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This review paper summarizes process safety aspects of abnormal situations and incidents in industrial facilities. Major accidents and their negative impacts on health, safety and environment have shown the importance of enhancing safety culture, understanding causes of abnormal situations and determine effective management strategies. Flaring is a safety industry practice to control and manage processes during normal and abnormal operations. This paper identifies current efforts made by industry and researchers toward better management of abnormal situation for flare reductions. These efforts came about because of current and impending environmental legislation. The paper finds that future strategies developed for ASM must consider safety as well as environmental and economic implications; and it highlights the challenges and future guidelines for safer abnormal situation management.

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Processsafetyandabnormalsituation

Major incidents in chemical industries have significant impacts in terms of economic losses, environmental emissions and in some cases human loss. Furthermore, they attract unwanted attention from monitoring regulating agencies and the public community at large. Overall, incidents threaten the existence and growth of the process industries. Process safety is a must to prevent such incidents form occurring. The literature indicates the key safety aspects necessary to prevent major accidents in process industries are: the creation of paradigm-enhancing organizations [1], establishment of process safety culture within the organization [2], consideration of broader social and cultural aspects of major accidents, awareness of total cost of major accidents [3], the development of inherently safe designs [4], dynamic operational risk management [5], and process safety competency [6].

According to AIChE center for chemical process safety (CCPS) leading and lagging metrics, almost every major incidents is the ultimate results of minor incidents, near misses or unsafe behavior, which are all indicators of some sort of abnormal situations or deviations from expected operations and practices [7]. The management of these deviations is termed abnormal situation management (ASM); and it relies on early warning systems, and timely diagnosis of their causal origins. Early diagnosis during operation under controlled conditions can reduce the loss of productivity during abnormal events. In cases where automated systems fail or are unable to cope, ASM provides decision support to operators to intervene autonomously and bring the process back to normal condi- tion [8,9,10]. That is one of the main reasons that human operators are necessary in processing facilities.

Abnormal situations can be caused by process failure, equipment failure, human/working contexterrors,or somecombinationofthethree.In most cases, it appears as a result of the interaction among multiple sources, which varies in their complexity and effect on the process [10,11]. Failure to detect abnormal condition, lack of understanding of its impact, lack of awareness of hazard and inappropriate response to handle abnormal situations are the sources of risk associatedwithexecutionfailures

under abnormal situation management [12,13].

Theultimateconsequenceofanabnormalsituation is not always explosions and fires. In cases where safety systems succeed in managing abnormal deviation, it can ultimately result in process shutdown.Ineithercases,theimpact resultsinlow production, poor product quality, reduced job satisfaction, schedule delays, equipment damage, violation of environmental releases, and other significant costs; in more serious cases abnormal situations lead to

industrial accidents that endanger human life [8,9,11,14].

MajorprocessaccidentsandHSEimpacts

Industrialaccidents

In last 40 years, many major accidents have taken place in industry. Table 1 presents selected major incidents that have taken place over the last five decades. The most notable the Bhopal accident is cited as one of the chemical industry's greatest tragedyandstillhasimpactsonthat community [1–3,5,6,15 ,16 ,17].Withallofitsnegative economic,environmentalandpersistentsocial impacts,the Bhopalincidentbroughtaboutthe establishmentofthe United StatesChemical SafetyandHazardInvestigation Board(CSB)in 1998[18].Thismarkedthefirstestablishmentof paradigm-enhancingorganizationsfollowedby severalotherssuchas:UnitedStates EnvironmentalProtectionAgency(EPA), NationalInstituteforOccupational Safetyand Health(NIOSH),Occupational Safetyand

Table1

Historyofprocessaccidents,theircausesandimpacts

Place	Year	Causalities/injury	Techno-economicandenvironmental impacts
Flixborough,UK	1June1974	Killed28peopleandinjured36	Vaporleakedfromrupturedpipework, Estimated cost:150millionUSD,Created colossalvapor cloudthatsoonfoundthe ignitionsource
Seveso,Italy	10July1976	Formedtoxiccloud,477peoplereportedskin injuries(burns andchloracne)	Reactoroverheatedresultingexplosion, Thecleanupcostwas110millionUSD Withindays atotal3300animalswerefound dead,
MexicoCity,Mexico	19November1984	killed600people,injured7000	Initial vapor cloud explosion was the start of a chainreactionofexplosions,therewere 19explosions
Bhopal,India	3December1984	Around2000peoplediedimmediately,with another8000 dyinglater	Isocyanategasleaked,UCCpaid 470millionUSDtosettlelitigationstemming from disaster
Chernobyl,Ukraine	26April1986	Killed30people	Resultofaflawedreactordesignthatwas operatedwithinadequatelytrained personnel. Thecostofdirectlossisabout15billion USD
PiperAlpha,UK	6July1988	Killed167workers	Operators were unaware due to a failure of the permit-to-work system that the pressure relief valve was not in place and that the blind flange wasnotproperly secured Thetotalinsuredlosswasabout3.4billion USD
Pasadena,USA	1989	Killed23people,injured130–300people	ExplosionoccurredatPhillips66complex
Toulouse,France	21September2001	Death-30,injured-10000	300tonsofammoniumnitrateexploded
Augusta,USA	2001	Killed3people	Thermaldecomposition
TexasCity,US	23March2005	Killed15,injured180	Aseriesofexplosionsoccurredwhena hydrocarbonisomerizationunitwas restarted andadistillationtowerflooded with hydrocarbons BPpaidmorethan1.6billionUSDto compensate victims
Buncefield,UK	2005	40peoplewereinjured	The incident occurred because a single piece of equipment failed (the level indicator) and there werenootherchecksin placetosafeguard
JilinCity,China	13November2005	Killed5andinjured70people	Initialinvestigationssuggestedthe explosion occurredafteroperators attemptedtounblocka nitrobenzene rectificationtower.Jilin'sBureauof ProductionSafetySupervisionand Administrationconcludedthatavalvewas left open,causingtemperaturestorise rapidly.
BPHorizon,USA	20April2010	Killed11workers,injured17	TheexplosioncausedtheDeepwater Horizontoburnandsink
Fukushima,Japan	11March2011	Officialfiguresshowthattherehavebeenwellover 1000deathsfrommaintainingtheevacuation	Following a major earthquake, a 15-meter tsunami disable the power supply and cooling of threeFukushimaDaiichireactors
WestTexas,USA	17April2013	Fatallyinjured12volunteerfirefighters,two membersofthe publicandcausedhundredsof injuries	Amassiveexplosionatafertilizerstorage and distributionfacility

Health Administration (OSHA). The EU named their industrial safety regulations as the Seveso Directivewhich isnamedaftertheSevesodisaster (1976). The Piper Alpha accident (1988) leads to radical change in regulations for offshore installationetc.Despitetheseactionsin1990s world continues to experience serious chemical accidents which create the urgency to reform the process safety management regulations. More emphasize was given on preventing the occurrence of major chemical accidents through saferdesignandquickresponsestoemergencies.

These organizations and regulations brought up new laws to increase chemical emergency preparedness and obliged the companies to develop process safety and risk management programs. They also investigate chemical accidents and recommend measures to prevent such catastrophicaccidents.

These accidents especially Flixborough, Seveso, and Bhopal have also shown the importance of reporting abnormal events in the chemical industries to collect accident precursors; that inspires different organizations and institutes to take required efforts. Wharton Risk Management Center has initiated to report near-misses, with near-miss management audits for better management of abnormal situations [19]. In addition, the AIChE center for chemical process safety (CCPS) has facilitated the development of a process safety incident database (PSID) to collect and share incident information, permitting industrial participants access to the database [20]. The Mary Kay O'Connor Process Safety Center at Texas A&M University (TAMU) has been gathering incident data in the chemical industries [21]. To record accidents, European industries submit their data to Major Accident-Reporting System (MARS) [22], while a database for chemical companies in the USA is created from risk management plans (RMP) submitted by facilities subject to Environmental Protection Agency's (EPA) chemical accidental release preventionandresponseregulations[22,23].

Apart from collecting necessary information regarding industrial accidents a number of research groups, safety related personnel, and environmentalists are continuously investigating/analyzing data of previous incidents to identify faults or remedies for similar scenarios. CSB or other incidents investigation organizations have frequently found that severe chemical accidents in the past occurred due to failure of timely management of abnormal situation. Another cause is the mishandling of process upsets/abnormal situation. Some such examples are: radioactive gas release in Three Mile Island-USA (1979) and ChernobylUkraine (1986), refinery fire and explosion in Grangemount-UK (1987), Milford Haven-UK (1994) and Texas City-USA (2005), gas plant explosion in LongfordAustralia (1998), thermal decompositioninAugusta-USA (2001), heat exchanger rupture in Houston-USA (2008)

[9,24 ,25 ,26–28]. Further, industrial statistics show that major catastrophes and disasters from chemical plant failure may be infrequent; however minor accidents are very common, occurring on a day to day basis, resulting in many occupational injuries, illness, environmental releases and

costing society billions of dollars every year [29 ,30].

Deepwater Horizon oil spill in Gulf of Mexico is a prime example of incalculable losses and sufferings due to abnormal situations. Researchers are still trying to understand the spill and its impact on marine life, the Gulf coast, and human communities. During the BP oil spill 1.4 million gallons of various chemical dispersants were injected straight into the Macondo wellhead, in order to reduce the amount of oil that reached the ocean surface. Five years after the spill some scientists believe that injecting dispersants directly at the wellhead may not have done much to help reduce the size of the oil droplets. [31]. Hence, to reduce the risk of such accidents, there continuous need to develop and advance safer strategies for the management of abnormal situations.

In process industries, flaring systems are part of the process and they are designed to handle planned process start-up, shutdown and for the venting of associatedgases during normal operations [32,33 ,34]. In addition, flaring practice is a standard safety measure during abnormal operations (e.g., upset flaring and emergency flaring) and during emergency shutdown. These kinds of flaring events are generally known as non-routine flaring which are intermittent, infrequent and are the result of operating conditions that are outside normal steady state plant process and equipment operations. Upset flaring occurs when the operations are outside the normal operating conditions and flaring is required to aid in bringing the unit operations back under control. Emergency flaring occurs when safety controls within facility are enacted to depressurize equipment to avoid possible injury or property loss resulting from explosion, fire or catastrophic equipment failure. Emergency flaring is a design scenario with the objective to depressurize the facility as quickly and safely as possible. Human operators are mainly involved to control the flaring events using their expertise [8–10,14,28,35].

Economic, health and environmental impacts of abnormal situation

Abnormal situations or events brought an economic loss of 20 billion US dollars per year to petroleum and chemistry industries in USA [36]. In UK, abnormal situations cause 27 billion US dollars per year [37]. The cost is much more when similar situations in other industries such as pharmaceutical, desalination, power, specialty chemicals etc. are included (see Table 1).

Industrial flaring during abnormal situation is necessary, yet a contentious economic, environmental and social issue because it does not only waste potentially valuable source of energy, it also adds significant carbon emissions and other nontoxic materials to the atmosphere that have been linked to cause cancers, asthma, chronic bronchitis, blood disorders, and other diseases in human health [38,39]. There have been over 250 identified toxins released from flaring including carcinogens such as benzopyrene, benzene, carbon disulfide (CS_2), carbonyl sulfide (COS) and toluene; metals such as mercury, arsenic and chromium; sour gas with H_2S and SO_2; nitrogen oxides (NO_x); carbon dioxide (CO_2); and methane (CH_4) which contribute to the greenhouse gases [40]. These pollutants cause acidity, temperature increase, and influence on the immediate environment, particularly on human health and plant growth [41]. Emissions from industrial flares also contribute to climate change or global warming. The implications of global warming have been linked to the melting of ice and the rising of sea level, which can lead to flooding and tsunamis across the globe [32,42,43].

Although flaring is standard practice in industry for safety, industrial flaring and its effect on the environment, ecosystem and society have gained the attention of researchers, environmentalists and policy makers [44–46]. Numerous protocol, international agreement and steps such as the Kyoto protocol, the United Nations Environment Programme (UNEP), World Bank Global Gas Flaring Reduction (GGFR) Programme have been initiated to mitigate the impact of industrial flaring. This is essential to avoid dangerous anthropogenic interference with the climate system. The recent COP21 United Nations Conference on Climate Change in Paris launched initiatives to end routine gas flaring at oil production sites around the world by 2030. Some 45 governments, organizations and oil companies had signed up to the plan by the end of that Paris climate change meeting.

Current status of global gas flaring

In recent years, governments and companies have had success in reducing routine flare gas with significant investments; global gas flaring rates have remained largely stable over the past fifteen years in the range of 140–170 billion cubic meters (BCM). This is equivalent of 3.5–5% of global gas production and 300 million tons CO_2 emissions [10,39,47].

Gas flaring data are available through various agencies including: national and international oil companies; International Energy Agency (IEA); US Energy Information Administration (EIA); the World Bank; Organization of Oil Exporting Countries (OPEC) [41,48,49]. The same sources also report greenhouse gas emissions. Greenhouse gas emissions from various sources including gas flaring are also available through the web site of United Nations Framework Convention on Climate Change (UNFCCC) [50].

ASM strategies

Abnormal situation management approaches are highly related to competitive decision making factors such as capital investment, technology risks, domestic market and its infrastructure, and political environment, companies' strategies etc. Although, a general comparison of all alternatives is difficult to make due to trade-offs, a case-by-case analysis is commended for the best action selection.

Recent efforts to address abnormal situation management can be found in the literature. These approaches can be classified as four major categories: Legislation implementation, flare recovery, flare utilization and incentive for carbon tax savings [34,51–54]. Different countries have adopted varying strategies depending on their situations/ scope and have had success in flare reduction. In Table 2, we have provided selected efforts from literature. The concept of providing credits for carbon-reduction along with viable flare reduction solution can be the added incentives that push industries for adopting enhanced flare management, recovery and utilizationstrategies[54].

Flare gas mitigation and recovery offers some real and tangible benefits: (1) recovered flare gas can be reused in process i.e., in heater burners and boiler burners; (2) reduces the amount of natural gas purchased by the facility; (3) reduced operational costs gas can lead to a fairly short return on investment for the capital investments of the flare gas recovery/mitigation units. Several research groups, institutions and companies are working simultaneously to find feasible alternatives for flare reduction. Recent articles and reports show different approaches have been suggested for effective flare reduction and utilization from a process perspective such as: utilization of renewable and alternative energy, process efficiency improvement, modernization and conversion of old techniques, waste streams recovery, combining heat and power through process integration, and innovative design solutionsforflare gas utilization [42,43,53,55 ,56–65]. The proposed new technologies and industry best practices show promise for reducing or eliminating routine flaring by minimizing waste gas production from processing units or using waste gas in production instead of releasing it; they capture energy value that would have otherwise been wasted and minimize emissions of greenhouse gases and other air pollutants. Recent efforts have focused on the capture and utilization the non-routine flaring [10,14,66,67].

The consideration of safety in the developed strategies for flare mitigation is critical. As process components and units are added to existing facilities, the control and safety systems become more complex. The consideration in safety in design has been addressed recently in the literature [4]. Recall, that flaring itself was a designed safety response to deviating process operations. Here the consideration of dynamic operational risk management and design of inherently safer systems are key safety aspects that must be addressed[5].

ChallengesASMandsafety

The following techno-economic constraints may be the reasons for lack of implementation and the resistance of private sector to invest in gas flaring reduction/recovery projects: small volume of certain sources of associated gases; remote geographical location; complications in the engineering design due to variation of pressure and temperature of the gas due to uncertainty; wide range of gas composition proven challenging in terms of purification and integration into existing flare systems; cost of purification, transport and injection.

Most of the recent works on flare minimization and utilization are focused on routine flare gas. The new advances have not seen success in implementation mainly due to the previously mentioned challenges such as design complexities, uncertainties and high magnitude of the non-routine flaring events. The process safety issues during handling high volume/pressure of flare in small duration are also limitations faced while managing non-routine flaring. Hence, there is a significant portion of the industrial flaring is still untapped.

Flare systems are dynamic processes, which are designed to handle planned or sudden outage in a plant. Some of the safety design criteria that need to be included in strategies targeting the management of non-routine/abnormal flare through utilization are: thermal radiation issues, the toxicity of the gases being disposed, JouleThomspon effects during rapid depressurization, material failure due to sudden cooling, and the dynamic nature of recovering non-routine streams[47,69–71].

Conclusion

Process safety is a priority for the operation of industrial facilities. Such efforts in developing a safetyculture,

Table 2
Selected examples of different ASM strategies used by countries for flare reduction in recent years

Country	Approaches	Effects	References
Angola	Flare and Relief Modification (FARM) project	Upgrades and modifies the flare gas and relief systems on 14 off-shore facilities, processes offshore natural gas liquids, and produces LPG	[68]
Argentina	Flare gas utilization for power generation	Aero-derivative gas turbine burns 0.45 million cubic meters per day of previously flared low Btu gas to generate about 40 MW of power	[55]
France	Energy alternatives	Introduction of nuclear power was able to reduce the emission drastically	[42,56,57]
Kazakhstan	Flare gas re-injection	Prevented annually more than 49 million tons of CO_2 from being released into the atmosphere	[55]
Nigeria	Flare recovery and utilization	Between 2000 and 2010, SPDC claimed that their flaring quantity dropped by half	[42]
Norway	Legislation implementation, incentive and penalties	Reduced gas flaring and venting significantly	[68]
Qatar	Legislation implementation	Changed the acid gas flaring limit from 1% to 0.06%	[47]
Saudi Arabia	Flare recovery and utilization	reduce the amount of gas flared	[42,56]
Southeast Asia	by conserving the gas from the oil field to be vented or flared	Reduce GHG emissions by 2.65 million tons of CO_{2eq}	[68]
USA	Flare recovery and utilization	Able to reduce their emission level by 5.8% between 2008 and 2009	[42]

developing inherently safer systems, and dynamic risk have led to better and safer working environments. However, as the literature has shown, industrial accidents continue to occur, with devastating economic, environmental, social impacts, and detrimental human loss. The leading causes for these incidents can be traced back to abnormal events. The management of such abnormal situations (ASM) is taking a leading role as global initiatives from the Paris agreement call for more stringent legislation on environmental impacts. The development and advancement of flare management is required to address such targets. Although recent efforts and works have shown successful strategies for routine flare, more efforts need to focus on the management of non-routine flares, such as developing approaches that are validated using dynamic control systems; advancement of flare system designs and considering multi-flare networks to handle flare reuse. The developed methods must simultaneously address safety and environmental challenges.

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